Automatic reconstruction of 3D city models from historical maps

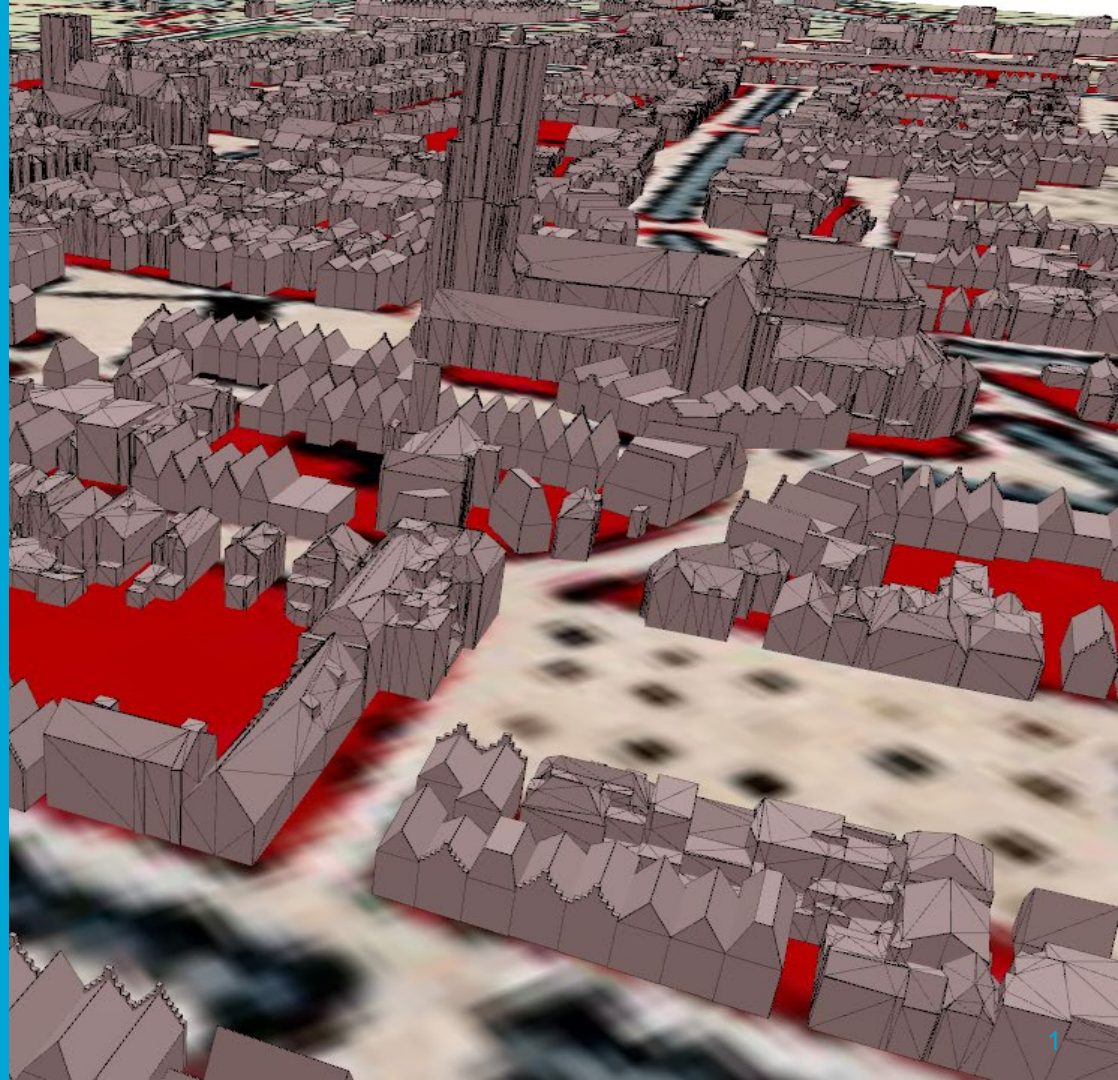
Camille Morlighem

June 2021

1st supervisor: Dr. Hugo Ledoux

2nd supervisor: Anna Labetski

TU Delft



Content

- Introduction
- Related work
- Methodology
- Results and analysis
- Conclusion

Content

- Introduction
- Related work
- Methodology
- Results and analysis
- Conclusion

TU Delft

Motivation

Historical maps?

Invaluable source of information…

about the pre-satellites era…

in many domains….

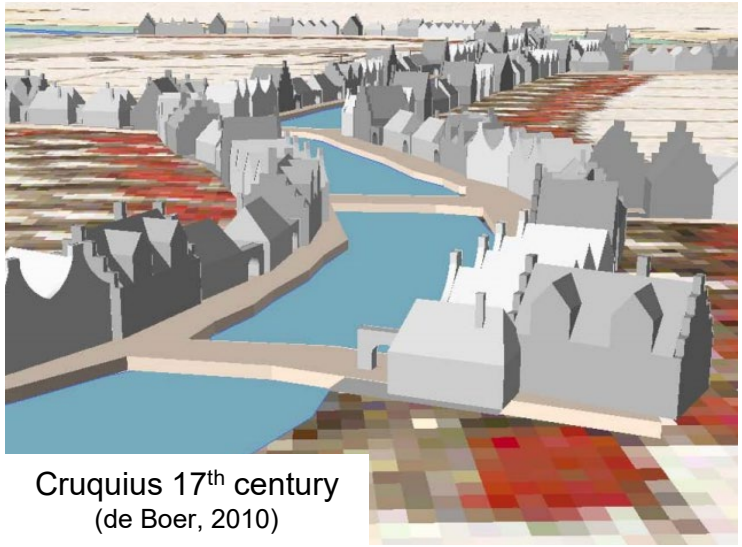
John Snow, London 1854

Source: McLeod [2000]

Motivation

Historical 3D city models?

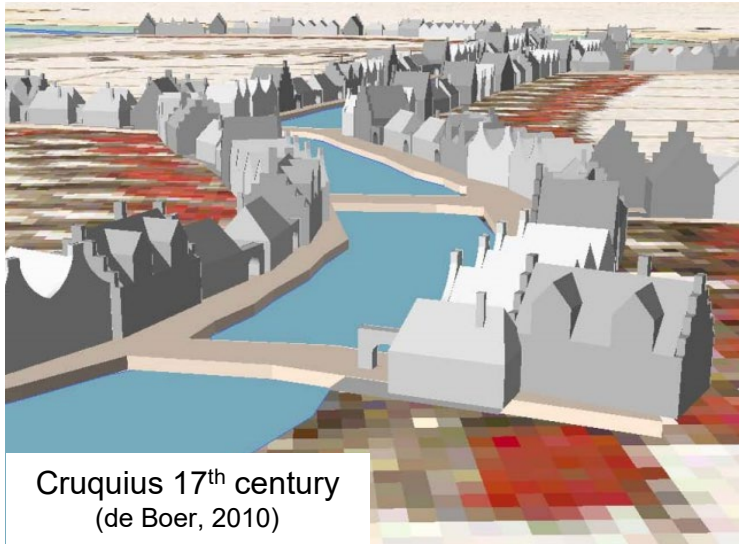
Rotterdam 1940



Cruquius 17th century
(de Boer, 2010)

Highly detailed VS. Simple

Research question



Cruquius 17th century
(de Boer, 2010)

"To what extent can be automated the process of reconstructing simple 3D city models from historical maps?"

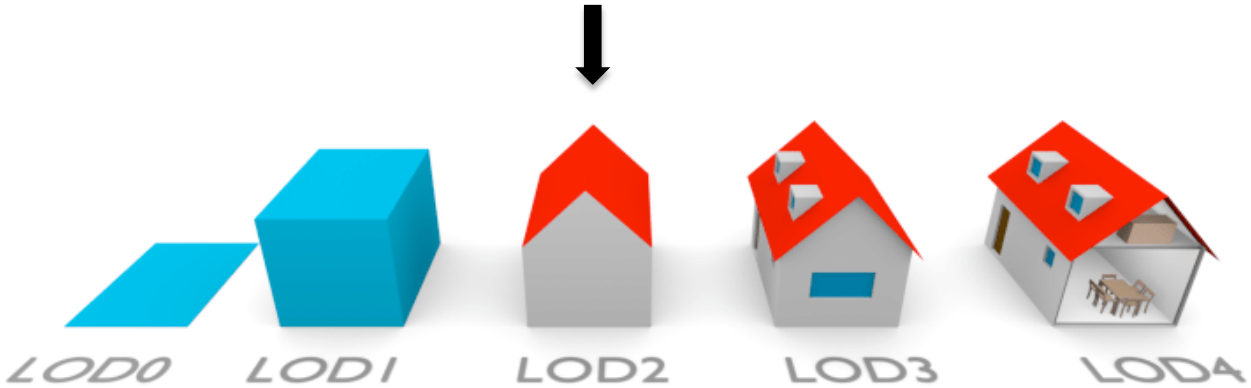
Objectives

1. Comparison of existing methods

Objectives

2. Automated methodology

- Models LoD2 buildings
- Maximises accuracy



Source: Biljecki et al. [2016]

Content

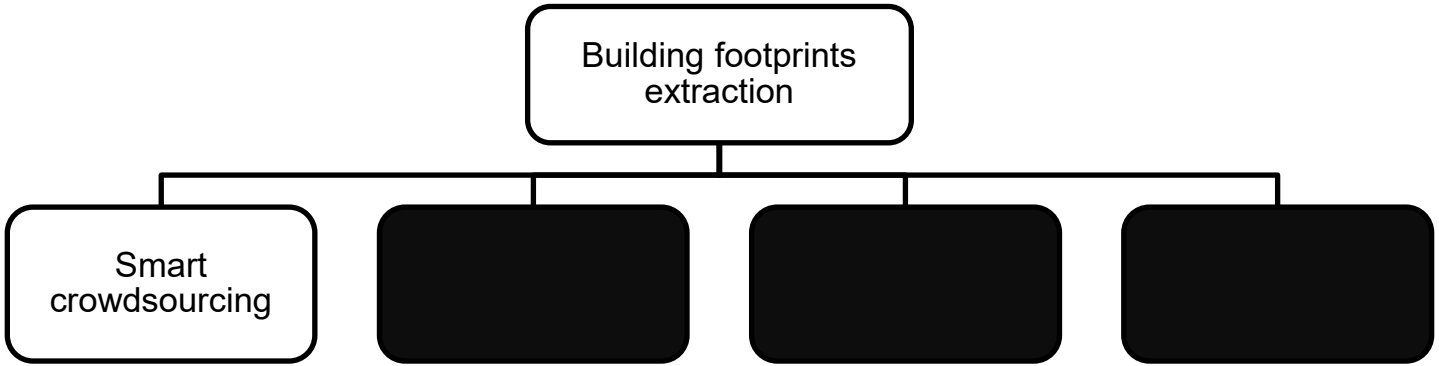
- Introduction
- Related work
- Methodology
- Results and analysis
- Conclusion

TU Delft

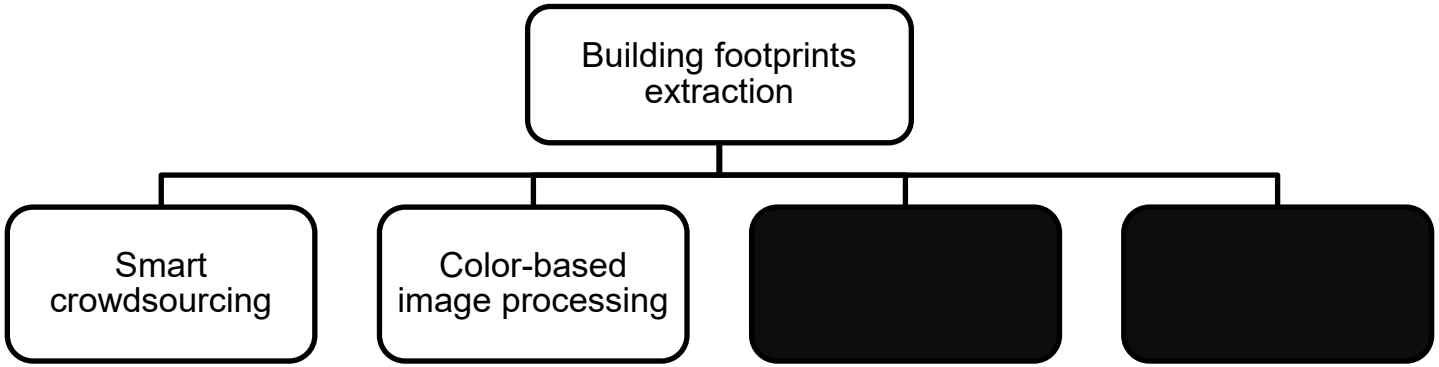
Related work

Building footprints extraction

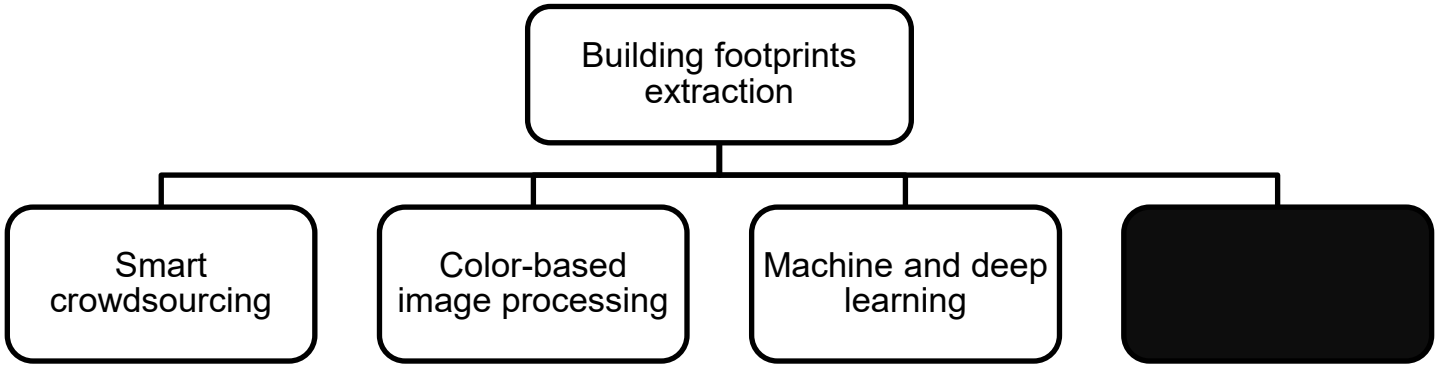
Related work



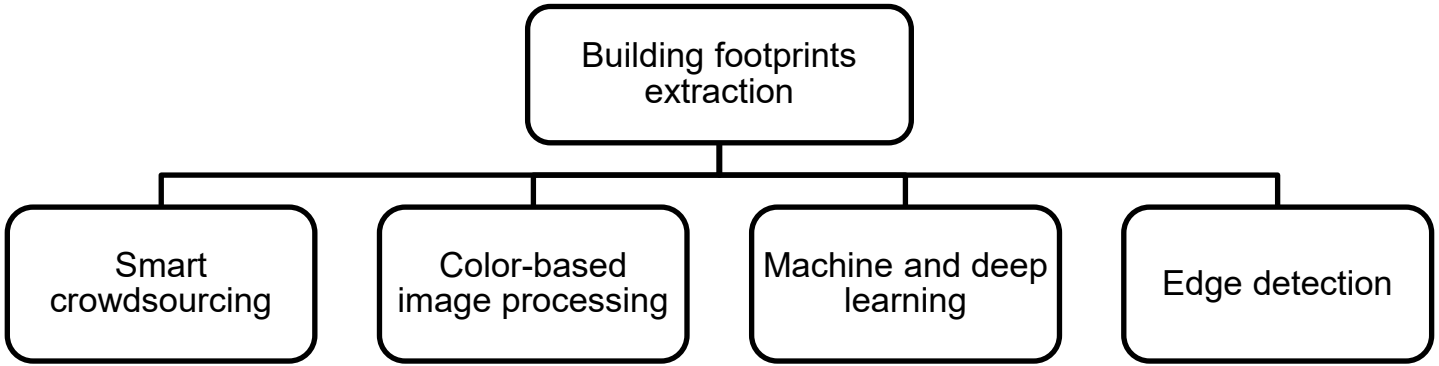
Related work



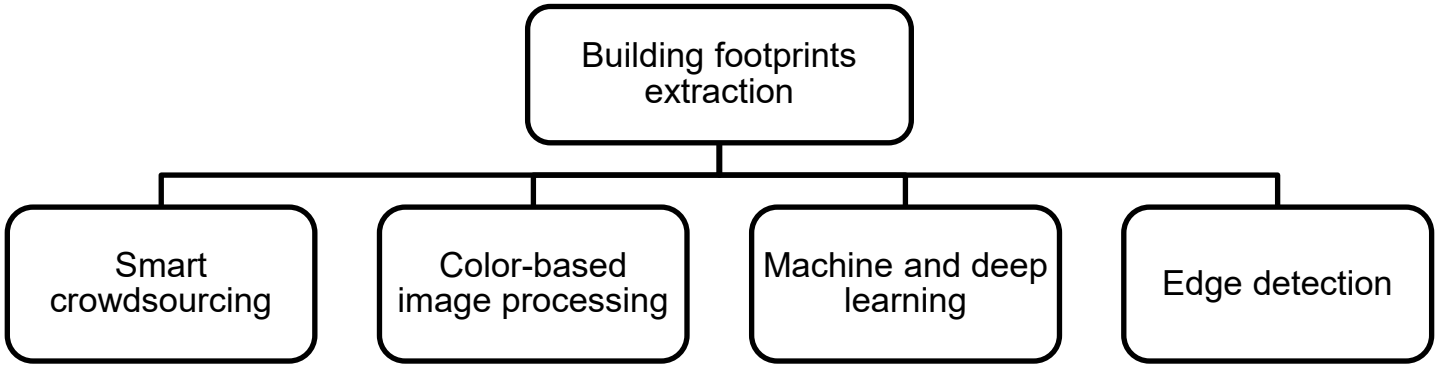
Related work



Related work

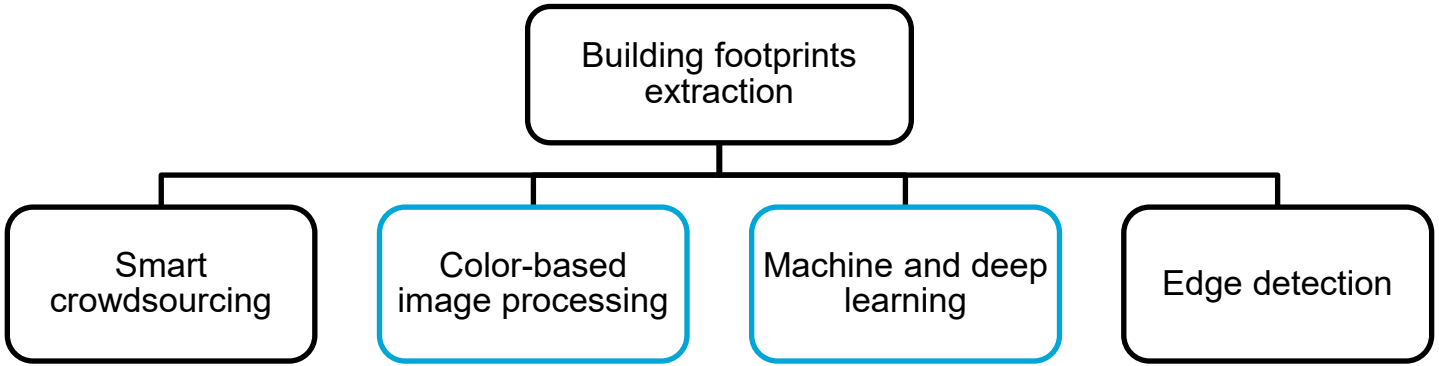


General observations



→ No perfect method

General observations



Content

- Introduction
- Related work
- Methodology
- Results and analysis
- Conclusion

TU Delft

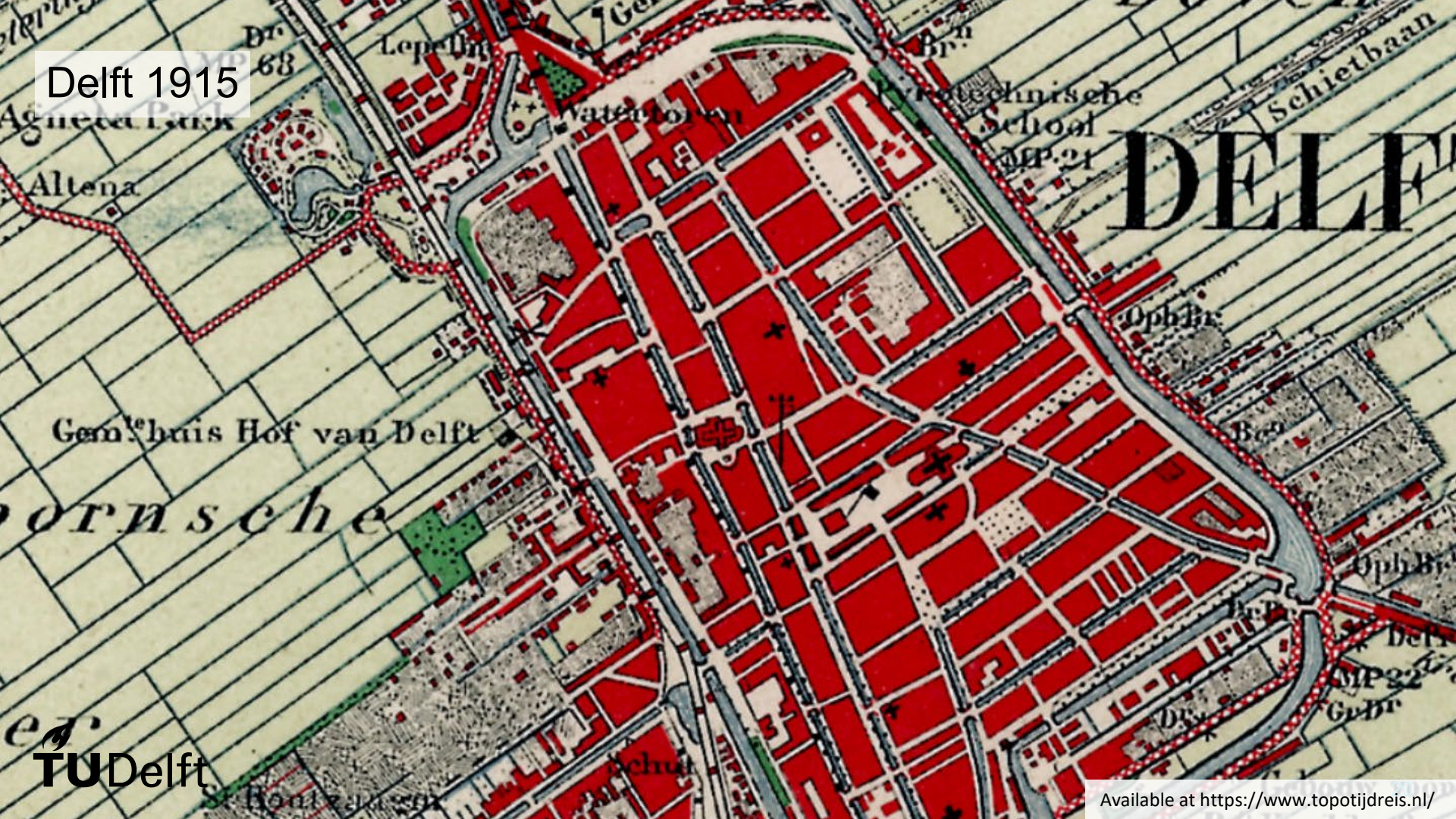
Case studies

- Delft

Implement the methodology

- Brussels

Test the application of the methodology for other study areas



Available at https://www.topotijdreis.nl/

Delft 1961

TU Delft

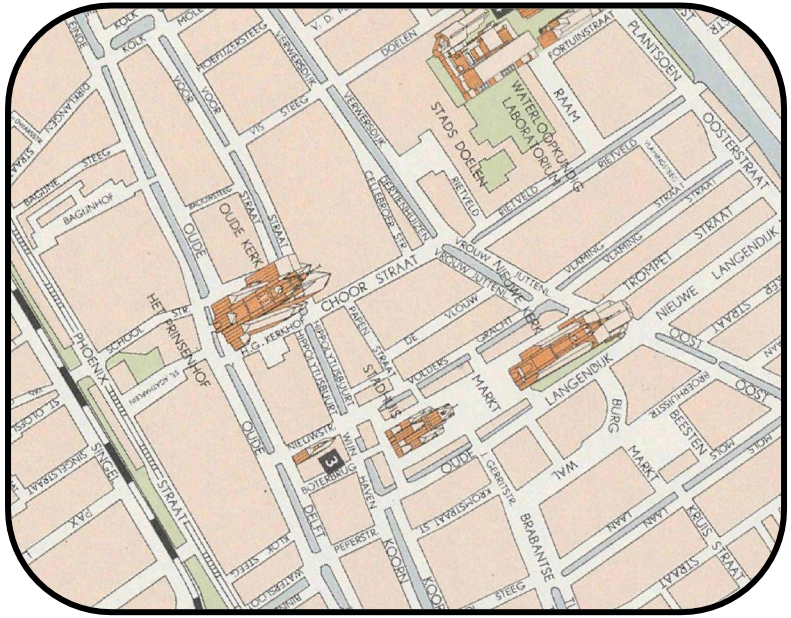
Available at https://heritage.tudelft.nl/nl/collections/lib-kaartenkamer

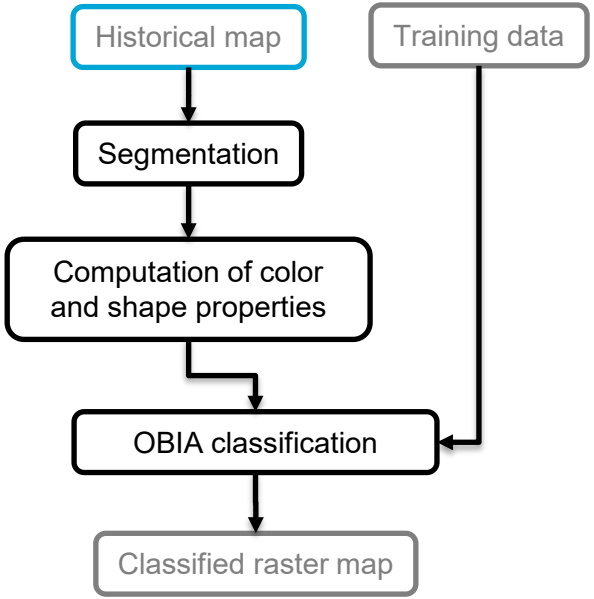
TU Delft

Available at https://gallica.bnf.fr

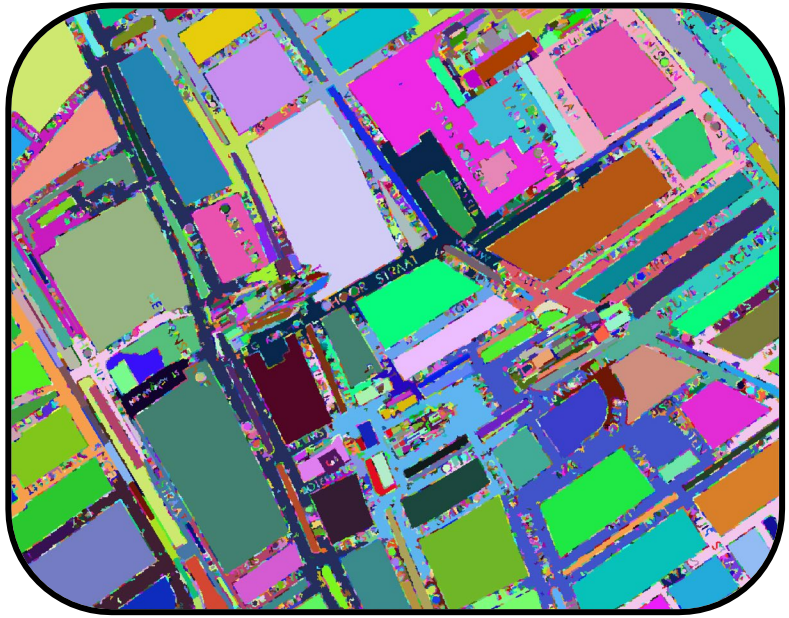
Brussels 1890
BOULEVARD DU JARDIN BOTANIQUE
BOULEVARD
Rue
MINISTÈRES
PARC
BOURSE
JARDIN BOTANIQUE
BISSCHOFFSHEIM
BOULEVARD DE L'ABATTOIR
BOULEVARD BARTHÉLEMI
CHAUSSÉE
TUDelft
Available at https://gallica.bnf.fr

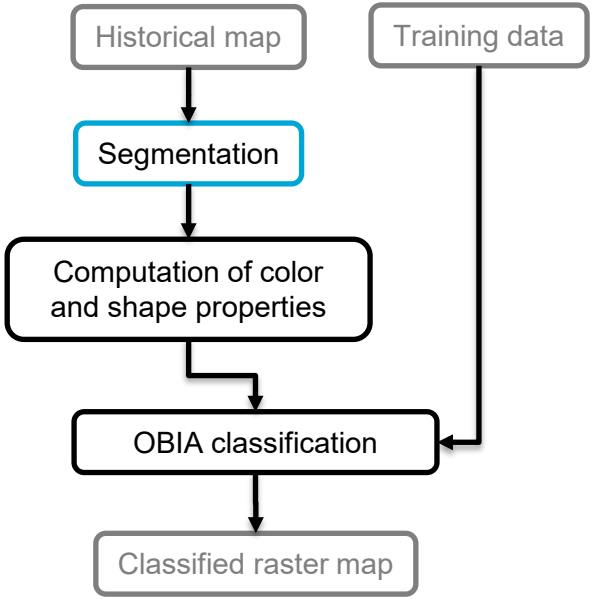
Building plots extraction



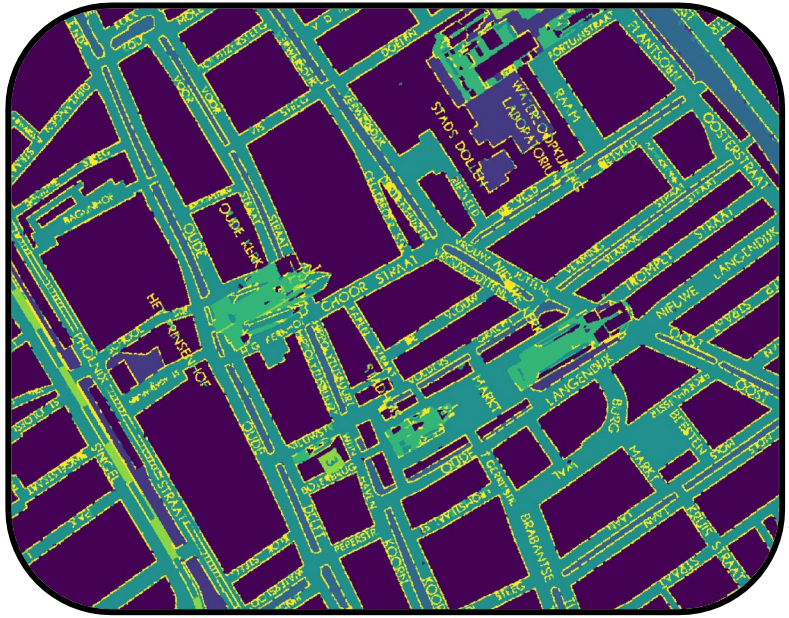


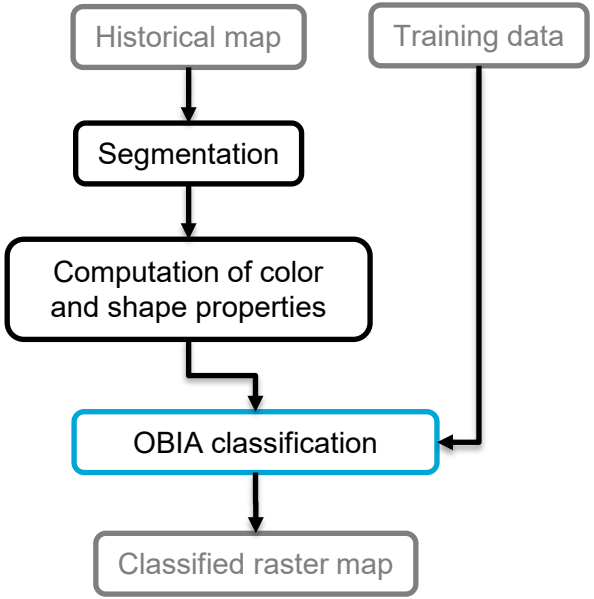
Building plots extraction



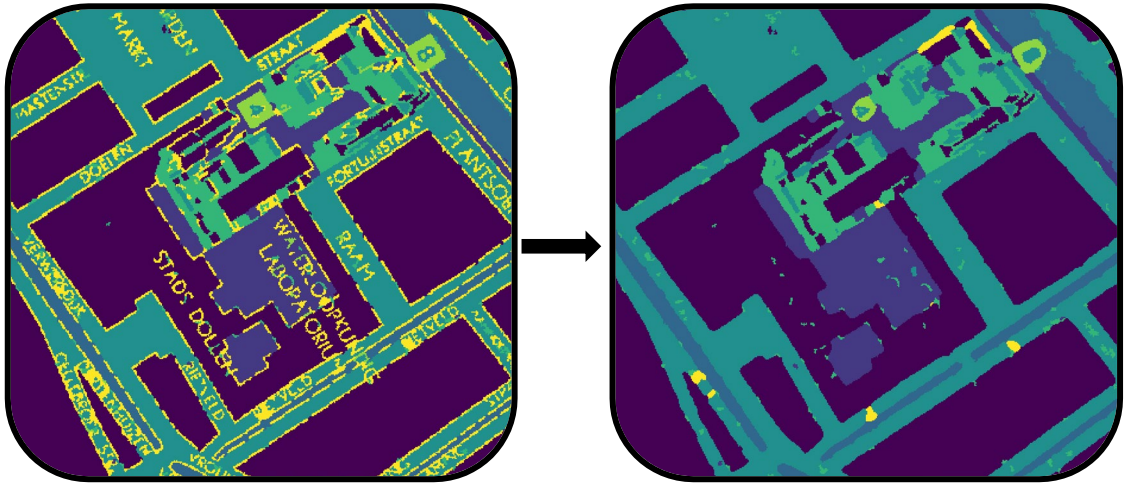


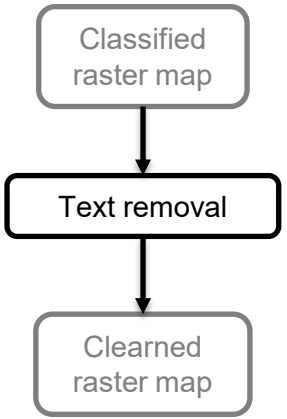
Building plots extraction



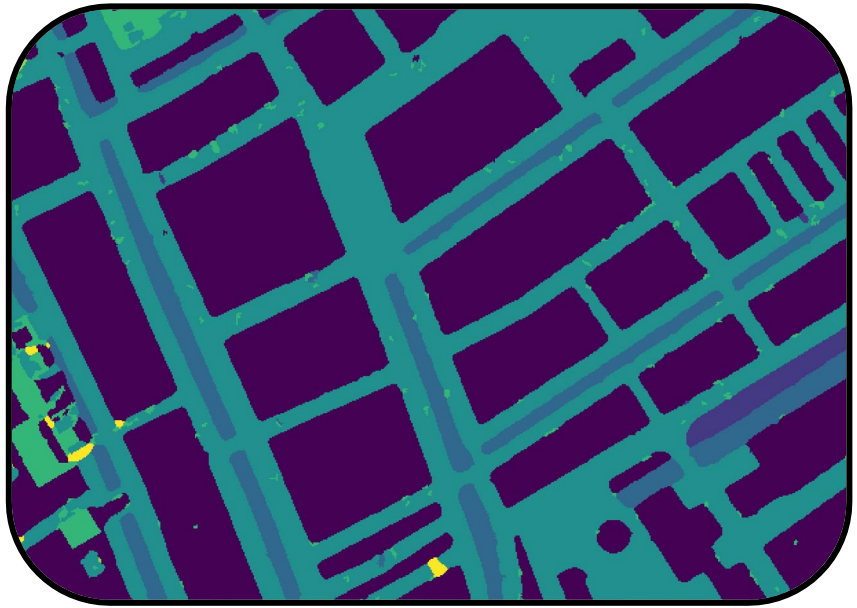


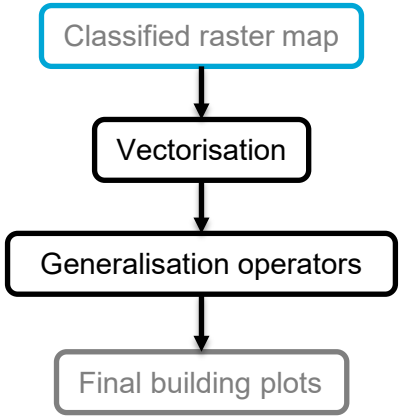
Building plots extraction



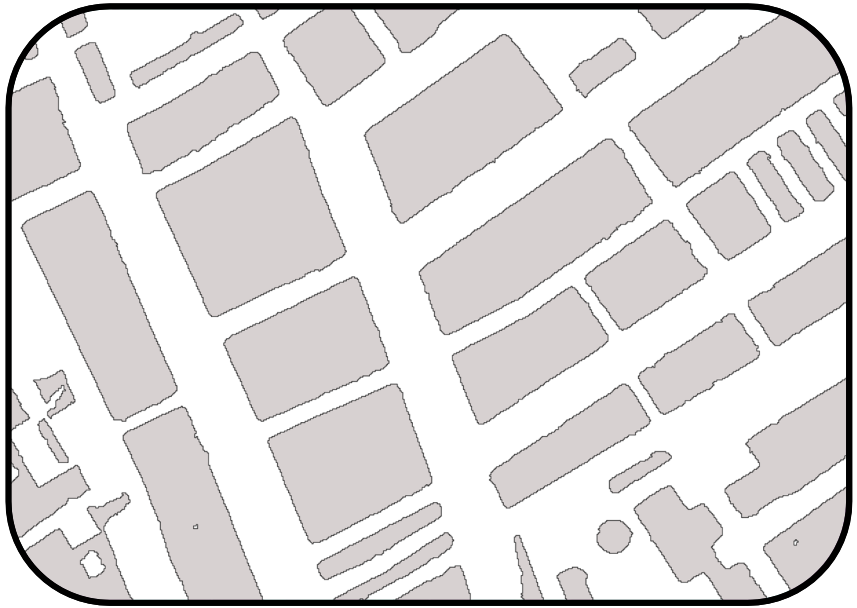


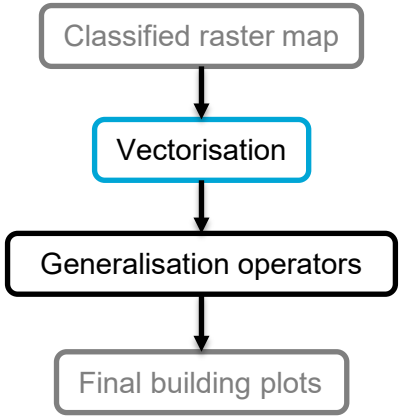
Building plots extraction



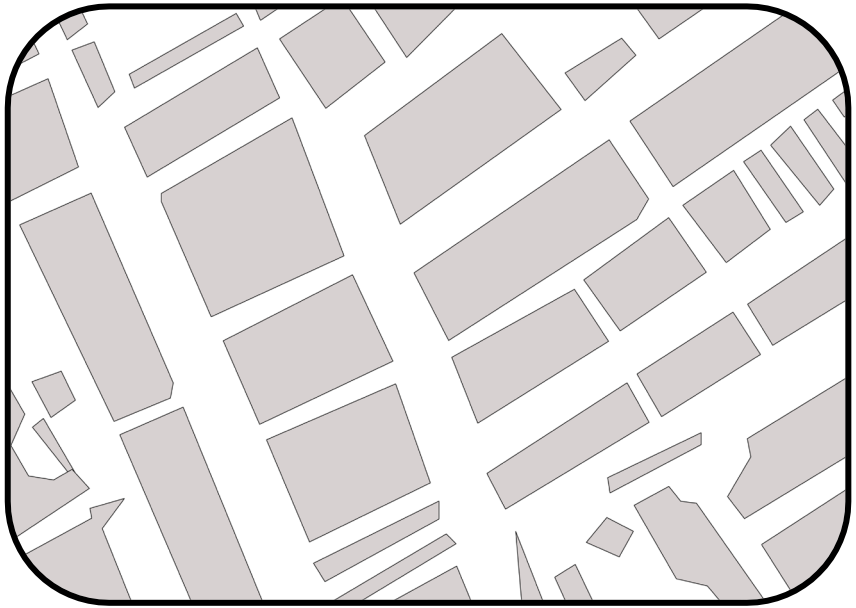


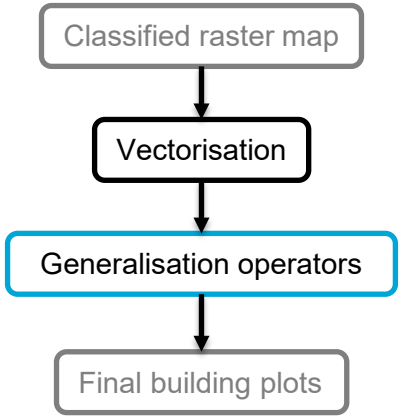
Building plots extraction





Building plots extraction





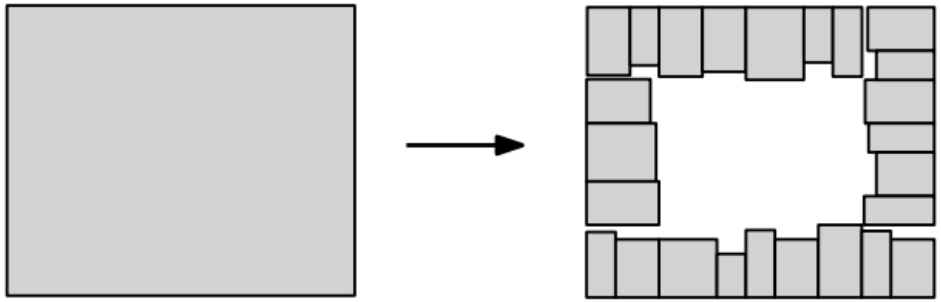
Reconstruction of individual footprints

1. Maps alignment

Historical building plots | Building footprints 2021 | Aligned entities

Reconstruction of individual footprints

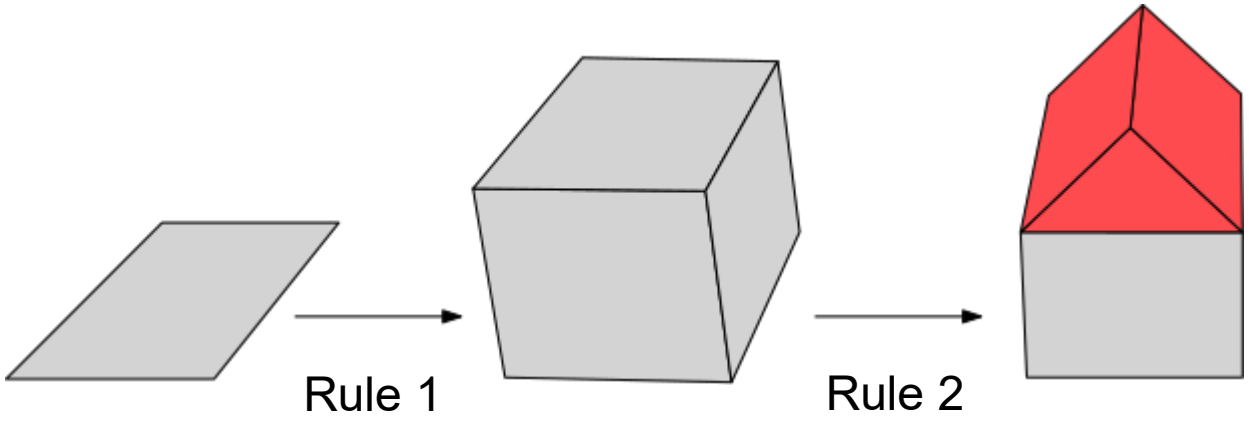
2. 2D procedural modelling



LoD2 buildings generation

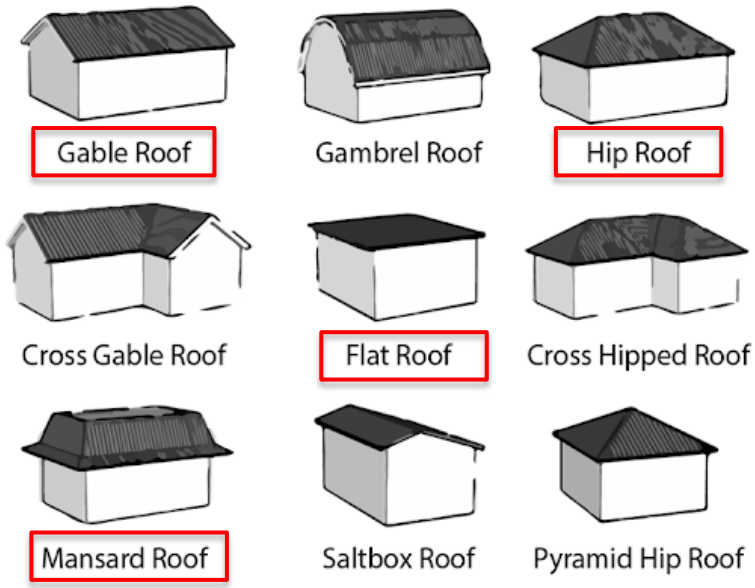
3D procedural modelling

Generate LoD2 buildings automatically from the building footprints using a set of rules



LoD2 buildings generation

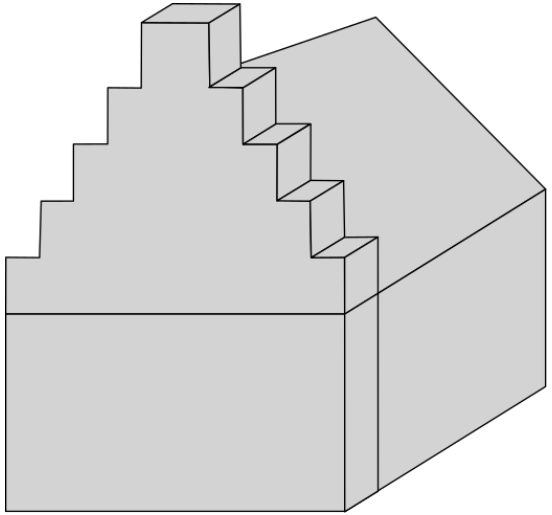
Blender addon



Source: Bondright Roofing Services [nd]

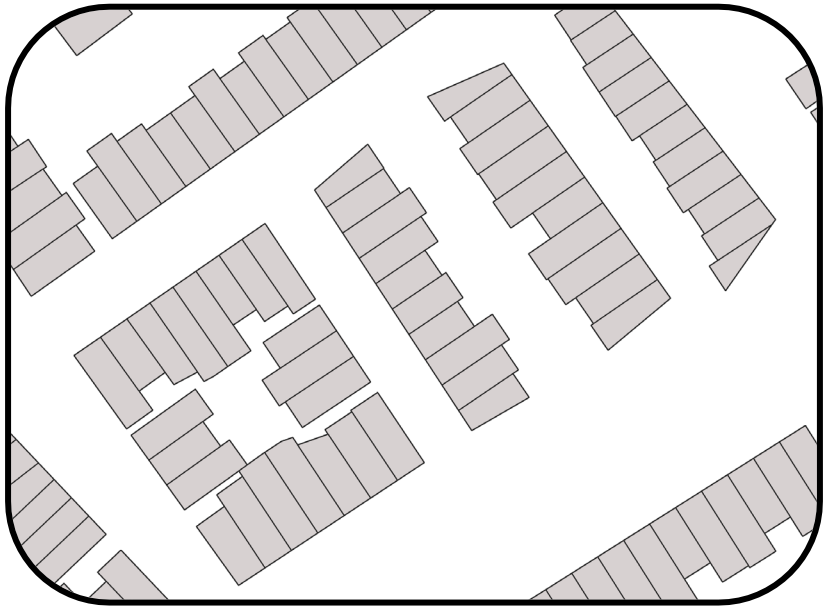
LoD2 buildings generation

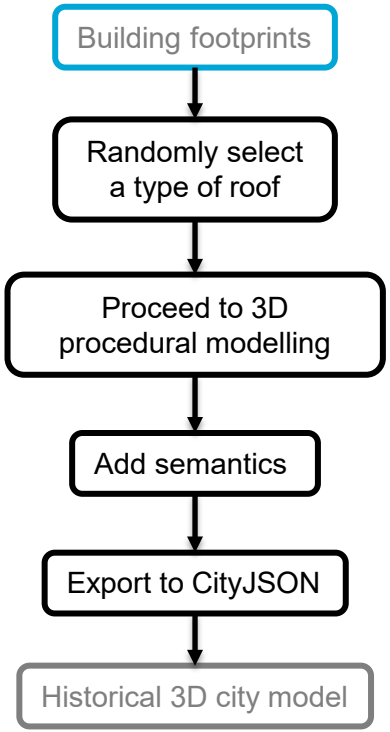
Crow-stepped gable roofs



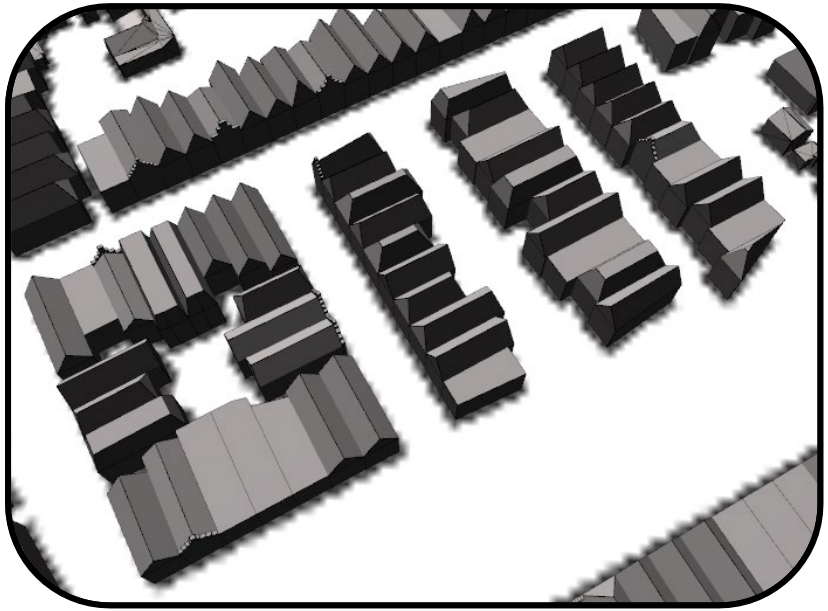
Source: Ryckaert [2012]

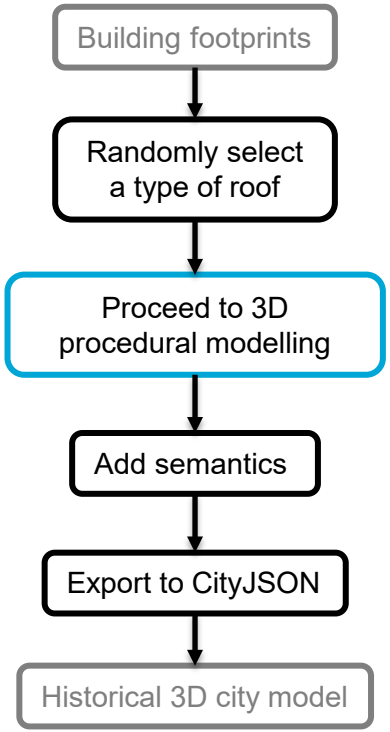
LoD2 buildings generation



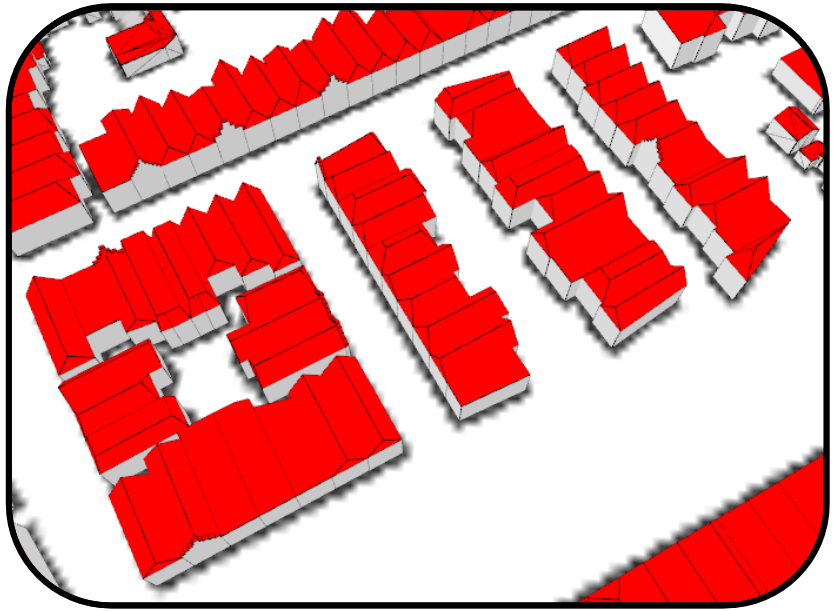


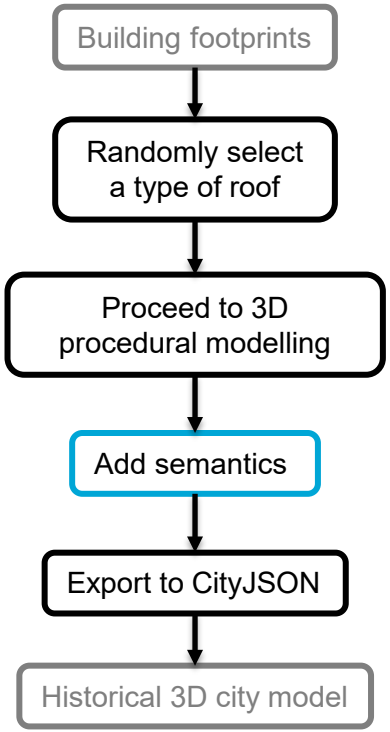
LoD2 buildings generation





LoD2 buildings generation





Content

- Introduction
- Related work
- Methodology
- Results and analysis
- Conclusion

TU Delft

Building plots extraction

Performance assessment

- Recall

Percentage of building plots properly identified in the ground truth

- Precision

Percentage of features classified as building plots and that are indeed building plots in the ground truth

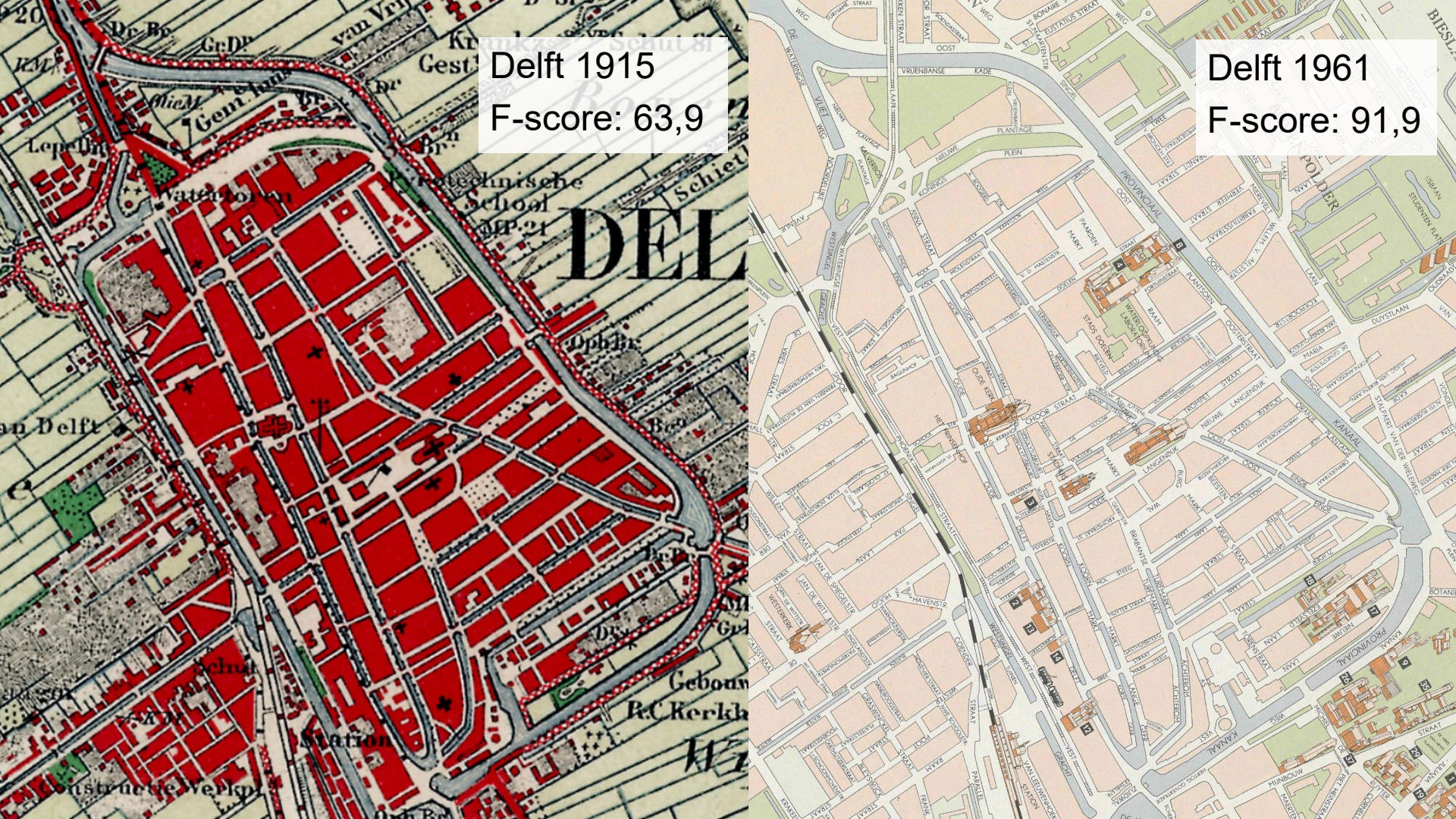
- F-score

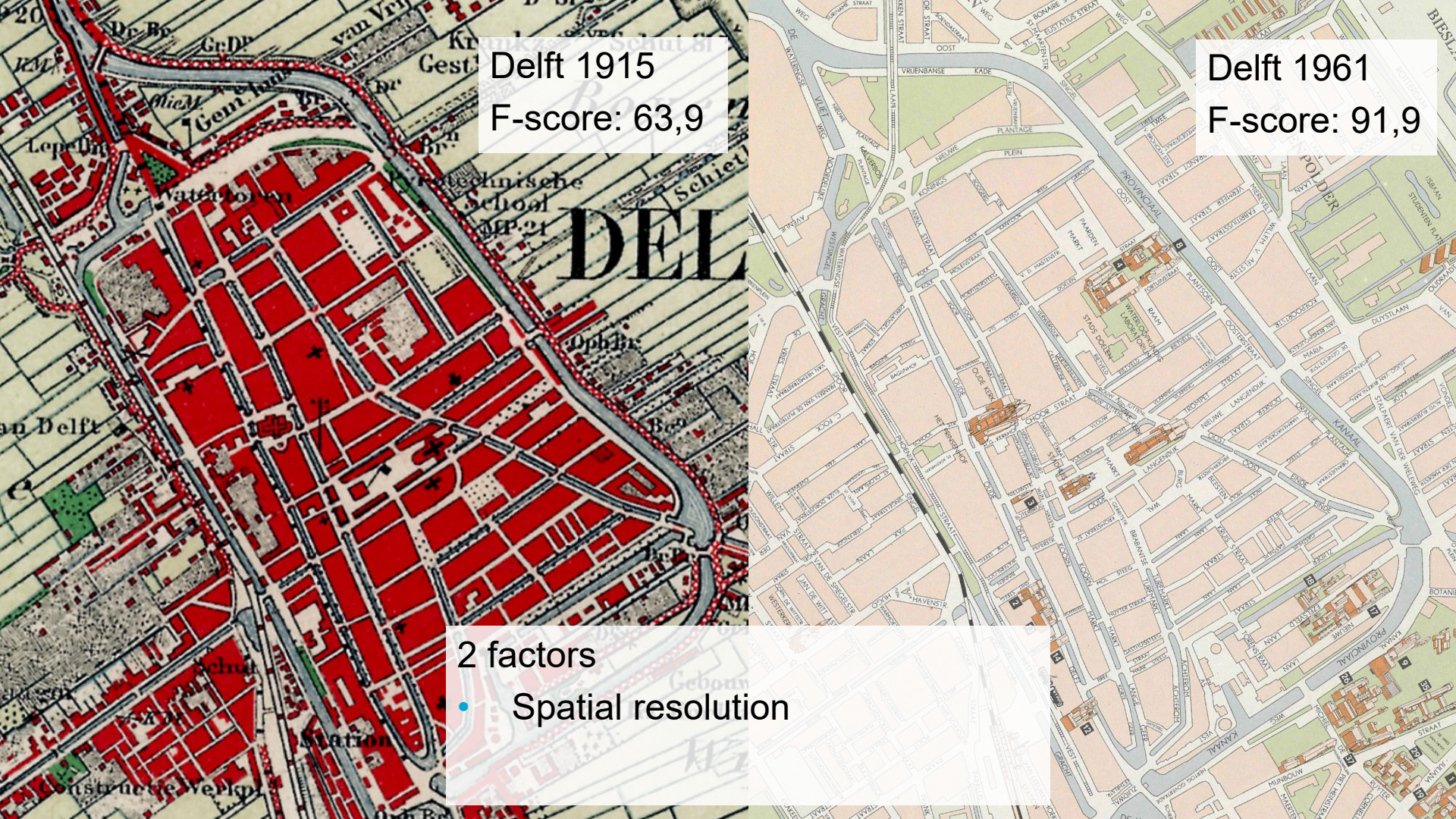
Function of the two metrics

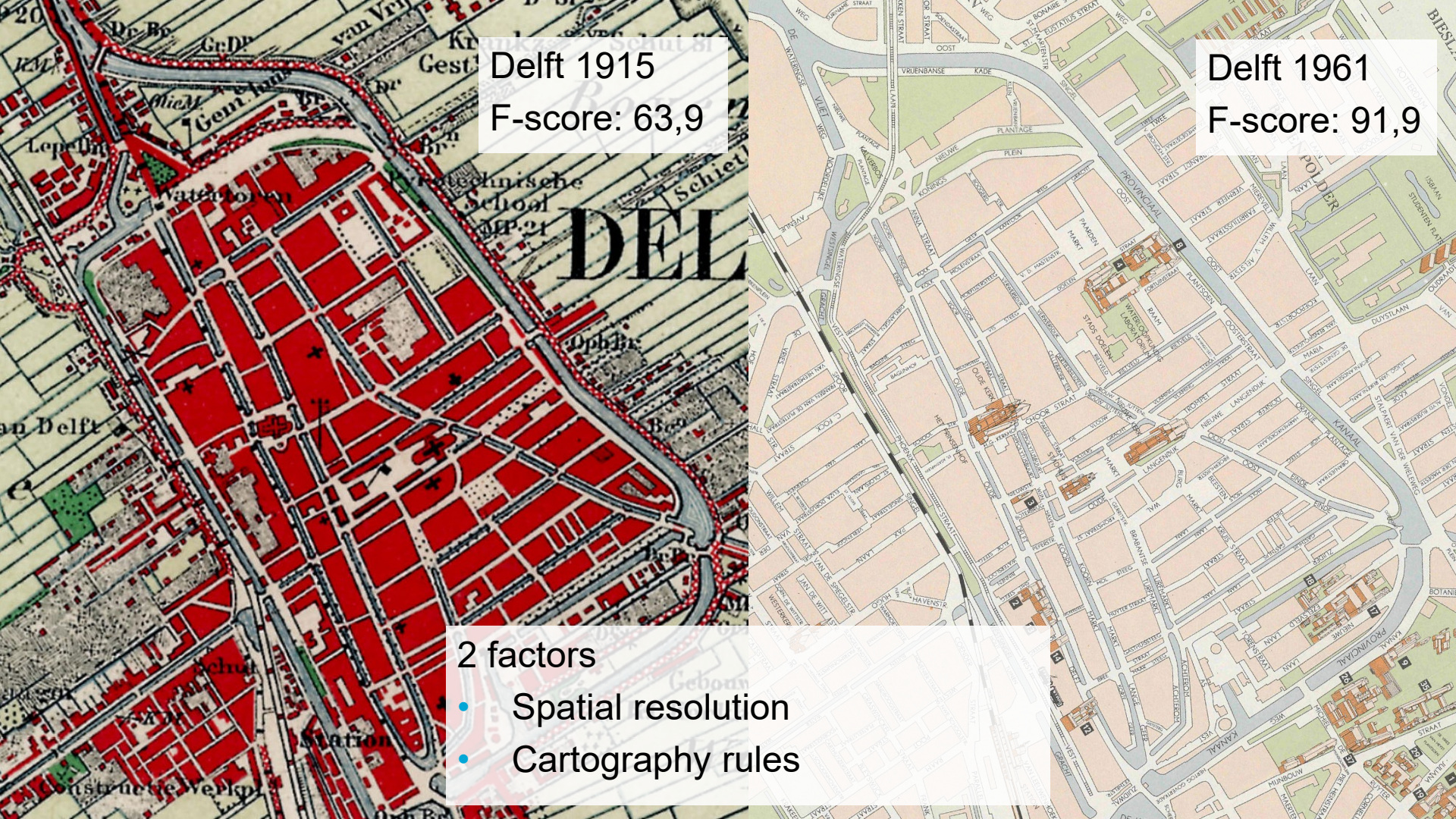
Building plots extraction

Historical maps	Precision	Recall	F-score
Delft 1880	60.7	67.4	63.9
Delft 1915	92.7	56.0	69.9
Delft 1961	93.6	88.9	91.2
Delft 1982	95.4	90.5	92.9
Brussels 1700	78.7	89.5	83.7
Brussels 1890	89.7	84.6	87.0
Brussels 1924	92.7	85.3	88.9

→ F-score > 83% for almost all maps


Delft 1915
F-score: 63,9
Delft 1961
F-score: 91,9


Delft 1915
F-score: 63,9
Delft 1961
F-score: 91,9
2 factors
Spatial resolution

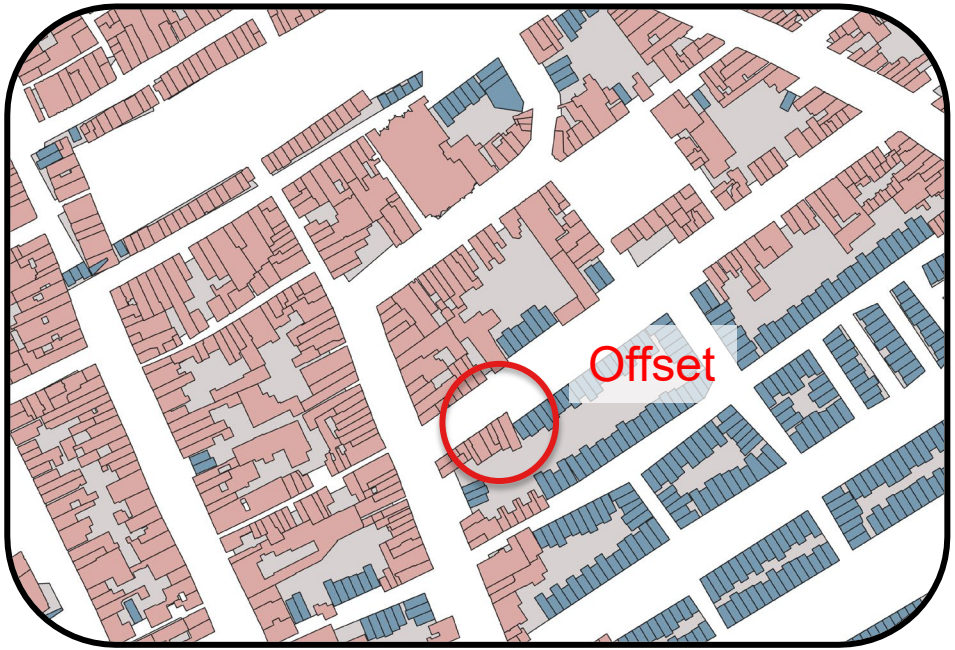

Delft 1915
F-score: 63,9
Delft 1961
F-score: 91,9
2 factors
Spatial resolution
Cartography rules

Reconstruction of individual footprints

Maps alignment + 2D procedural modelling

Reconstruction of individual footprints

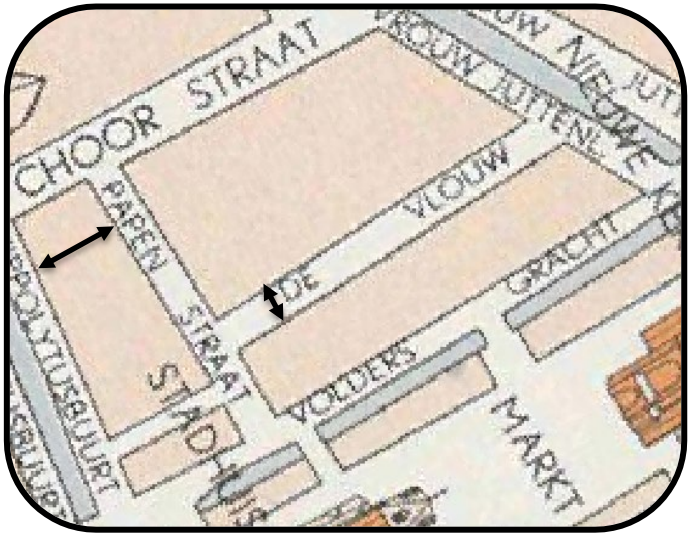
Maps alignment + 2D procedural modelling



Reconstruction of individual footprints

Inaccuracies?

Building plots 2021



Building plots 1961

LoD2 buildings generation

1. Delft

Delft 1880

TU Delft

Delft 1880

Delft 1915

TU Delft

Delft 1880

Delft 1915

Delft 1961

TU Delft

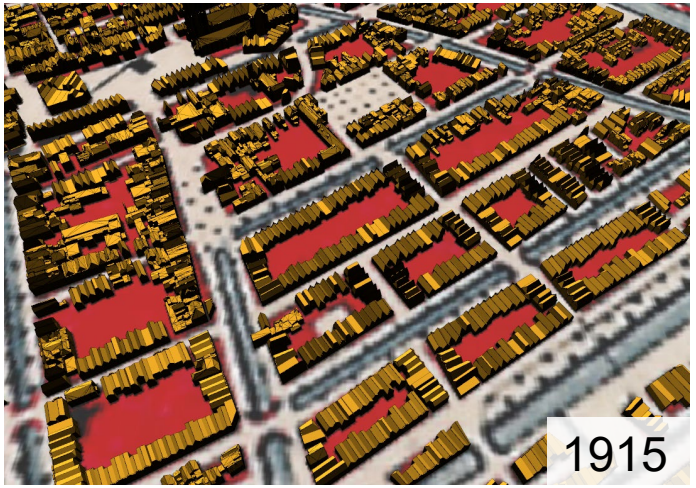
Delft 1880

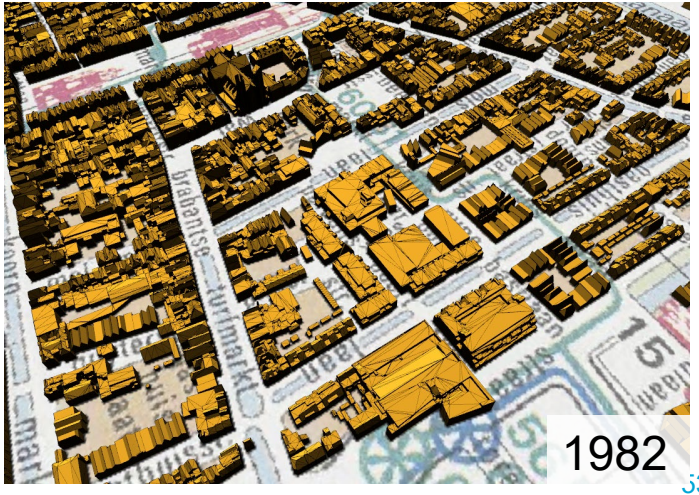
Delft 1915

Delft 1961

Delft 1982

TU Delft

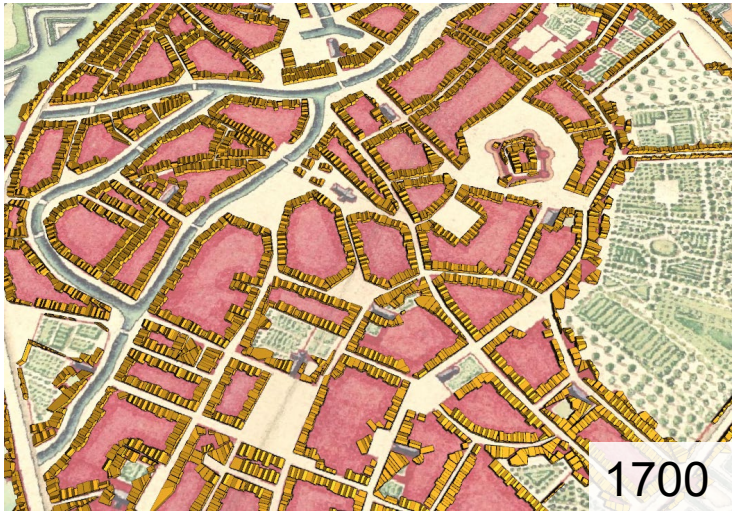




TU Delft

LoD2 buildings generation

2. Brussels


1700

1890

1924

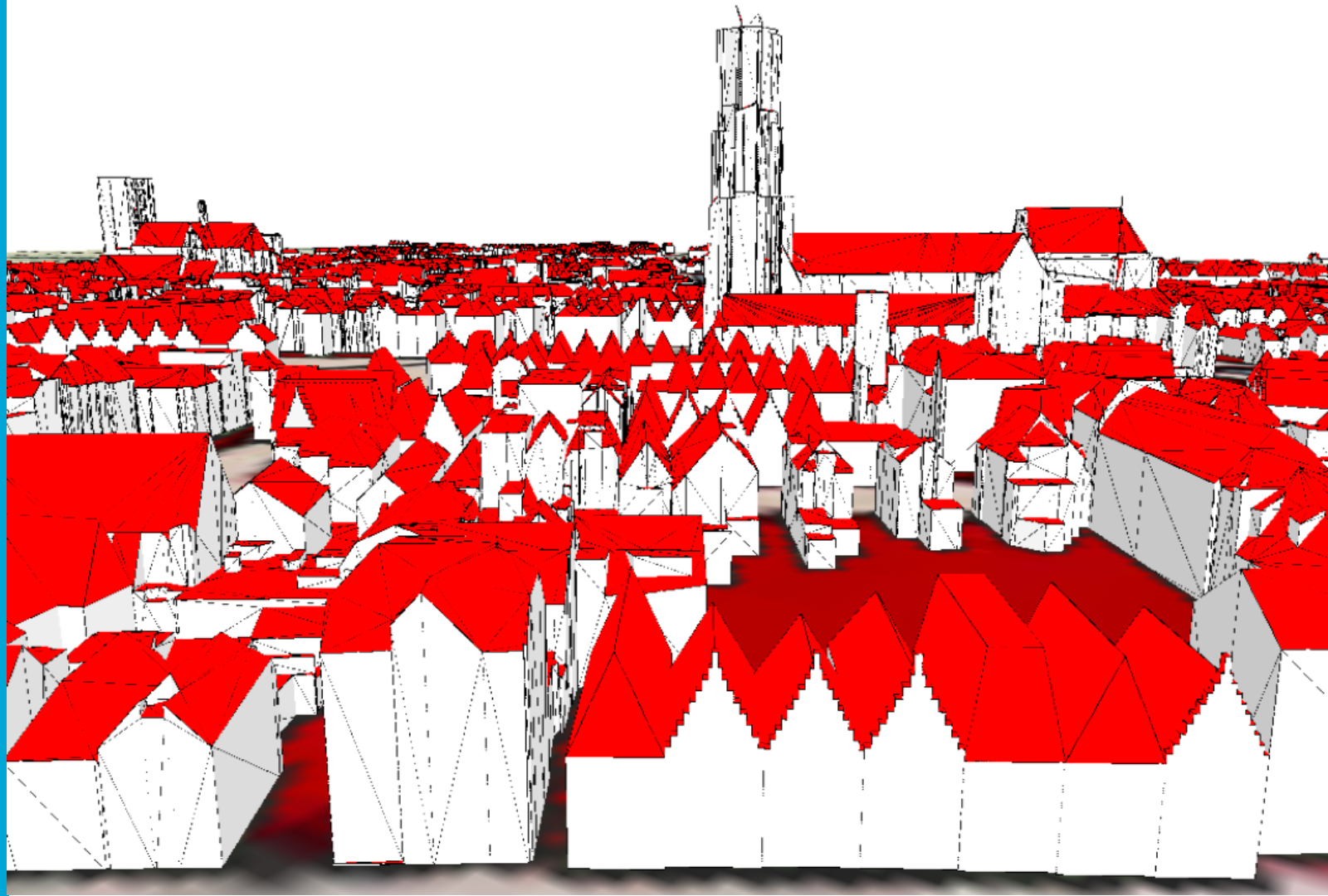
LoD2 buildings generation

Results assessment

- Visual assessment

Are the buildings generated as expected?

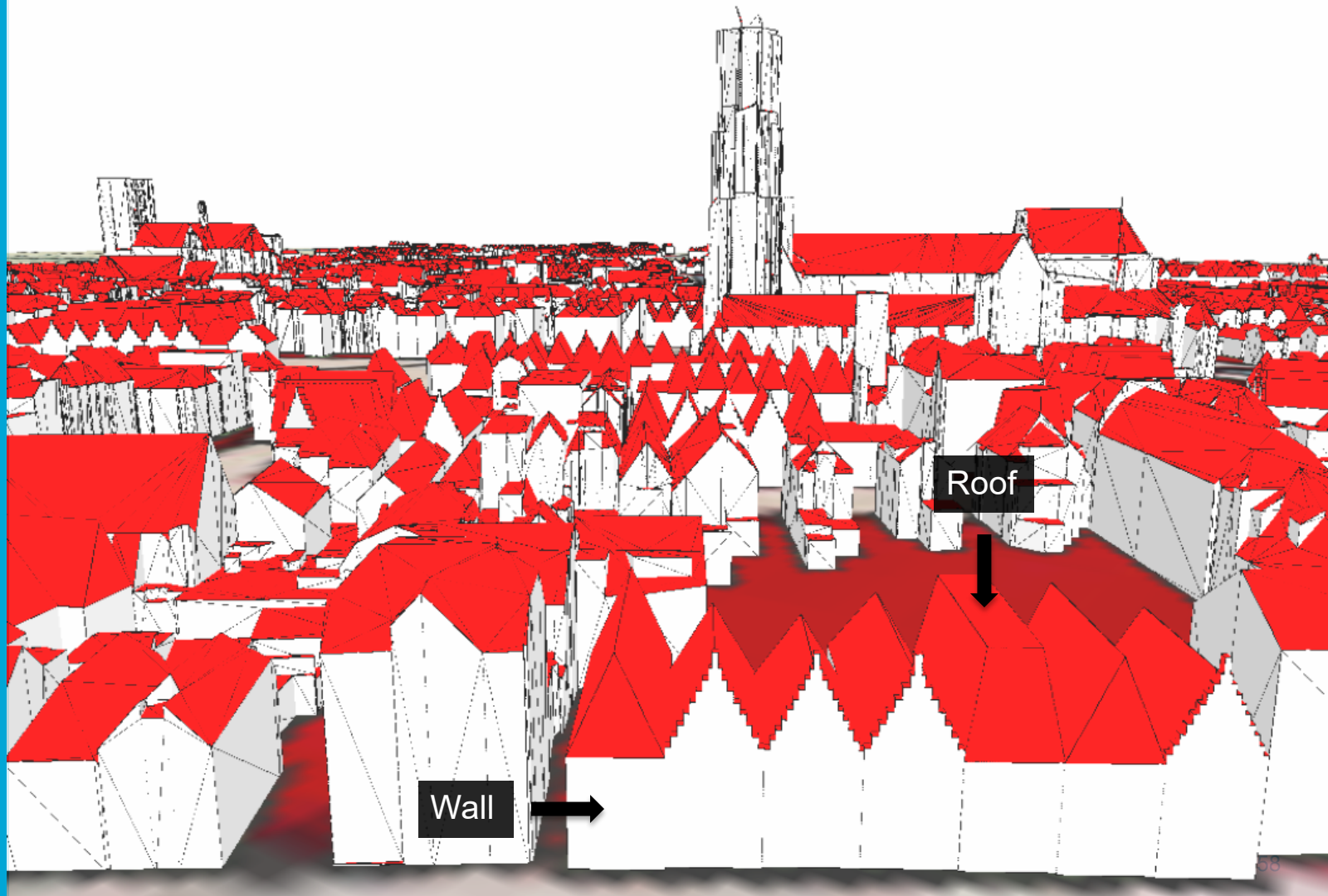
Delft 1915



TU Delft

Delft 1915

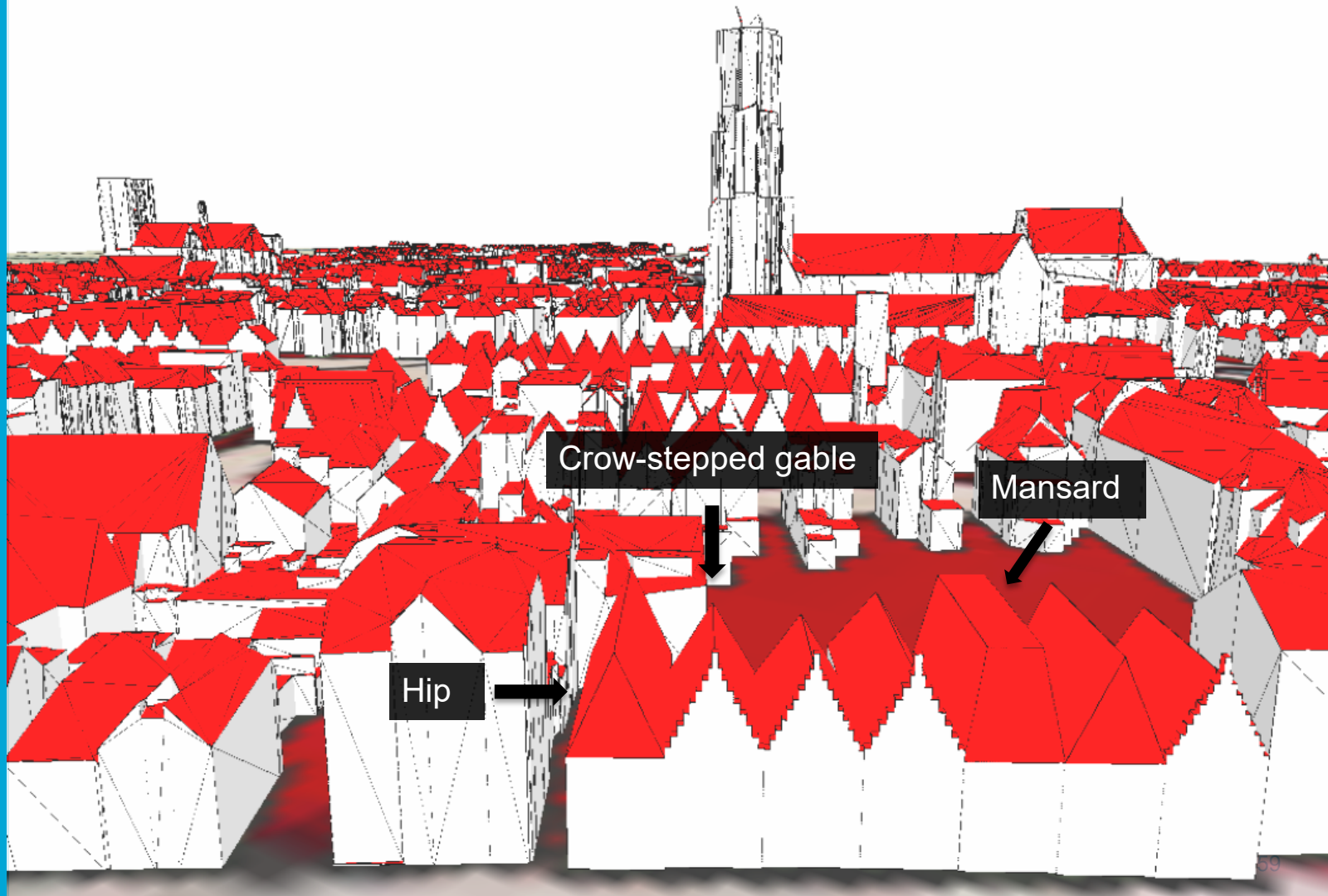
Semantics



TU Delft

Delft 1915

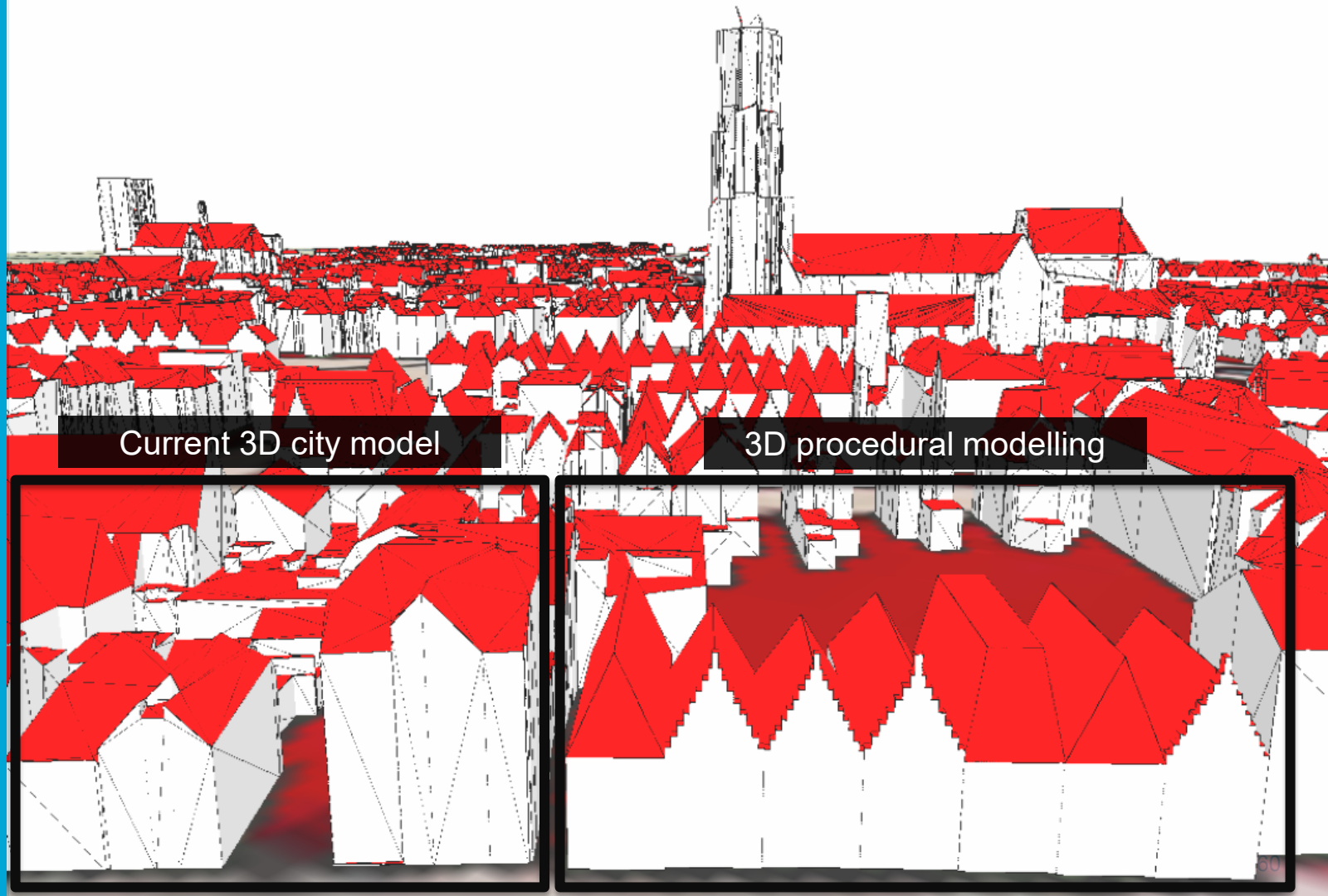
Semantics
Roof types



TU Delft

Delft 1915

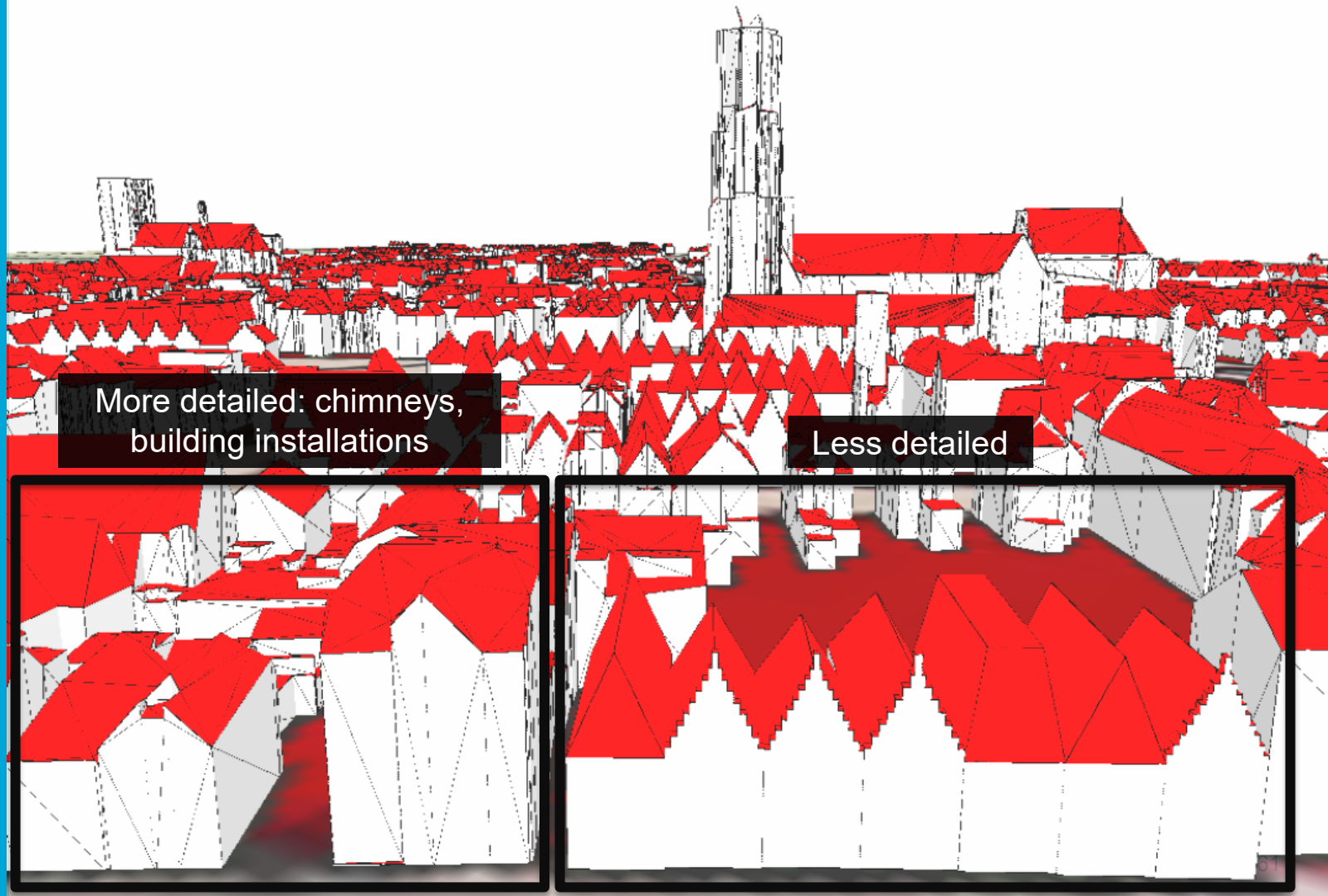
Semantics
Roof types
Multi-LoD



TU Delft

Delft 1915

Semantics
Roof types
Multi-LoD



TU Delft

LoD2 buildings generation

Results assessment

- Validity assessment

Is the CityJson file valid ?

→ Yes !

Are the geometries valid according to standards?

→ > 99% for all historical 3D city models

Content

- Introduction
- Related work
- Methodology
- Results and analysis
- Conclusion

Research question

"To what extent can be automated the process of reconstructing simple 3D city models from historical maps?"

- Limited user interventions
  - Provide input datasets
  - Create training data
  - Pass some user-defined parameters
- Methodology flexible
- Large processing time

TU Delft

Discussion

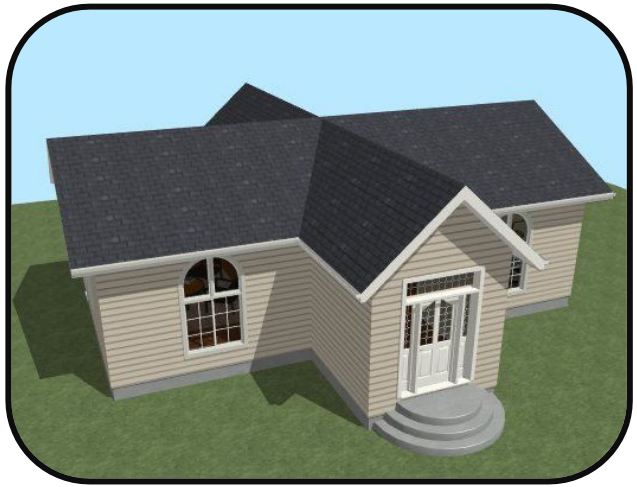
- Application to other study areas?

Discussion

- Application to other study areas?
- Use cases for such historical 3D city models?
  - Preservation of the cultural heritage
  - Pragmatic applications
  - New applications not investigated yet

Future work

- Addition of procedural modelling rules
- Use of additional historical sources
- 3D reconstruction of other features



Source: Home Designer® Software [2019]

Source: Jones [nd]

Thank you !

References

Biljecki, F., Ledoux, H., and Stoter, J. (2016c). An improved LOD specification for 3D building models. Computers, Environment and Urban Systems, 59:25–37

de Boer, A. (2010). Processing old maps and drawings to create virtual historic landscapes

Bondright Roofing Services (n.d.). BONDRIGHT ROOFING GUIDE TO RESIDENTIAL ROOFING TYPES. http://bondrightroofing.co.uk/bondright-roofing-guide-to-residential-roofingtypes.php

Home Designer® Software (2019). Creating a Cross Gable Roof. https://www.homedesignersoftware.com/support/article/KB-01043/creating-a-cross-gableroof.html

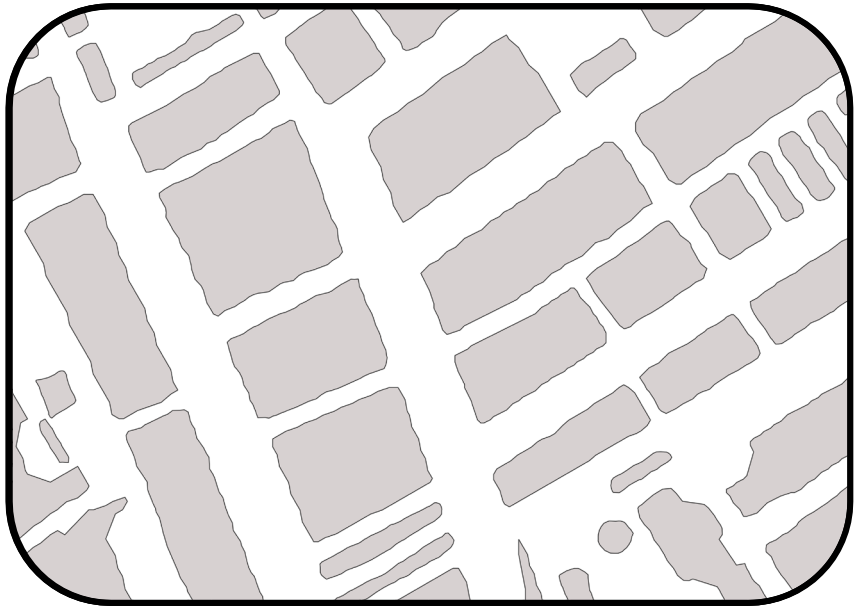
Jones, S. (n.d.). Leaning Houses in Amsterdam. https://thetravelbunny.com/canal-leaninghouses-amsterdam/

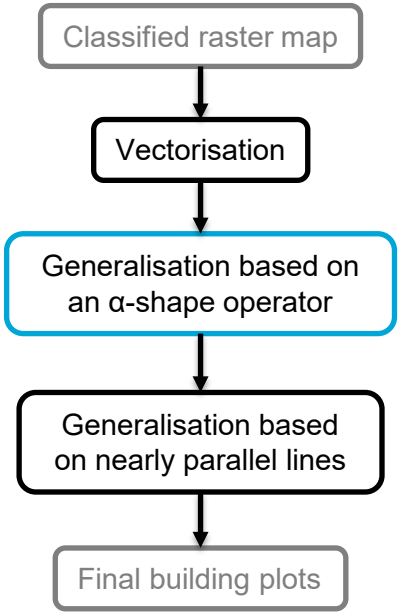
McLeod, K. S. (2000). Our sense of Snow: the myth of John Snow in medical geography. Social Science Medicine, 50(7):923 – 935

Ryckaert, M. (2012). Bruges (Belgium): houses at the Potterierei. https://commons.wikimedia.org/wiki/File:Brugge Houses Potterierei R02.jpg

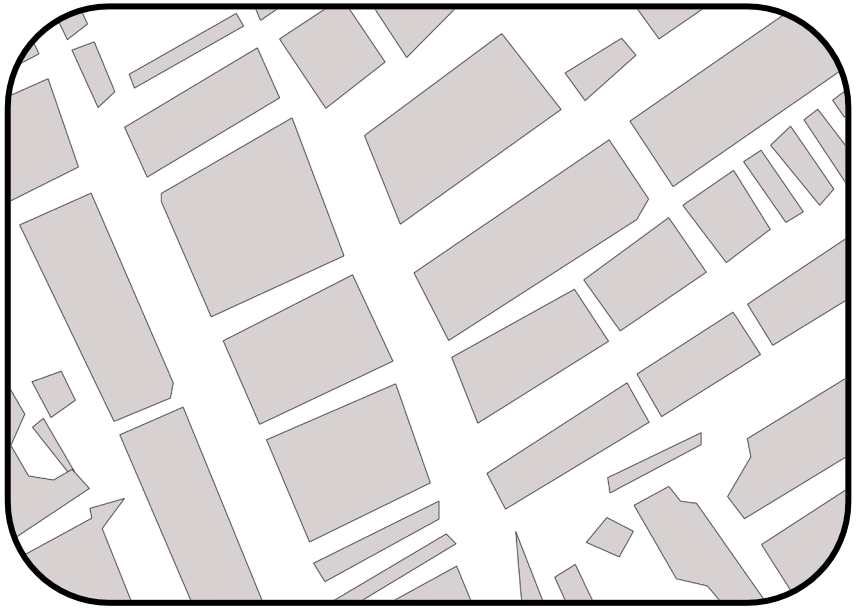
Extra slides

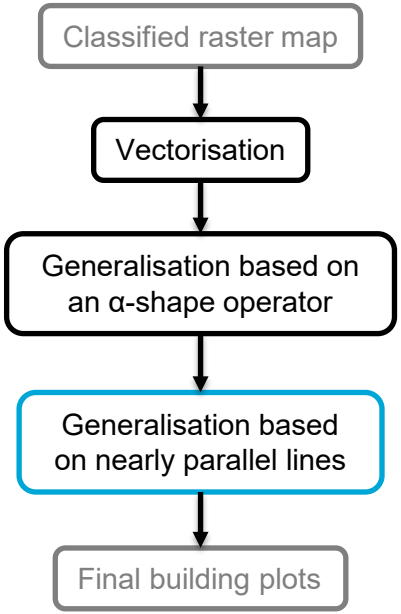
Building plots extraction





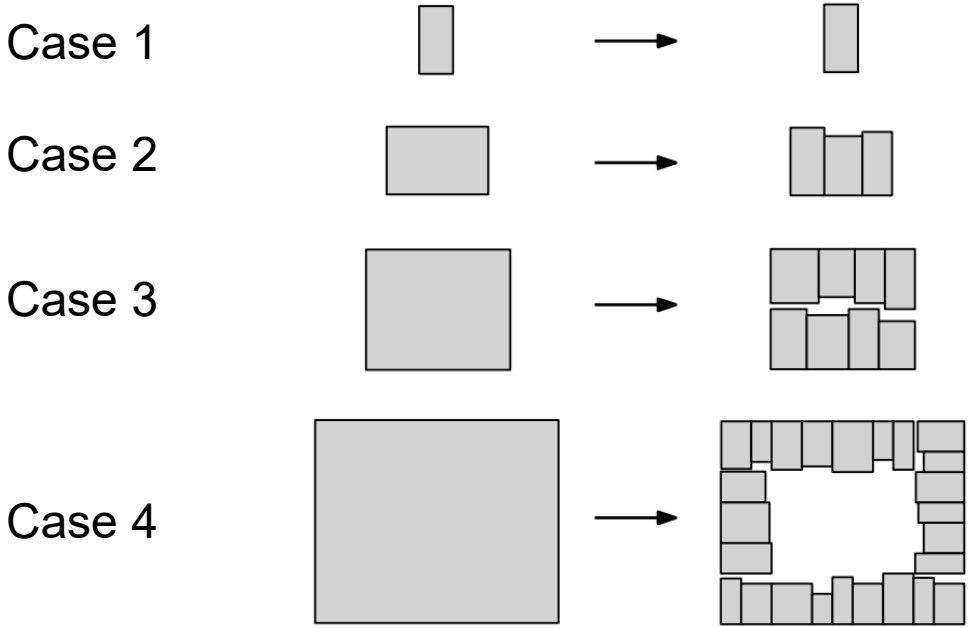
Building plots extraction





Reconstruction of individual footprints

2D procedural modelling



LoD2 buildings generation

Assign building reference heights

- Ground height

Use current elevation dataset (point cloud)

- Roof height

Use height attribute of the aligned building footprints

Contributions

- Literature review about historical 3D city modelling
- Comparisons of different methods
- Automated subdivision of building plots into footprints
- Blender addon for 3D procedural modelling enriched
- Automated reconstruction of historical 3D city models

TU Delft